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Mini Project Report on

**Comparison of Reinforcement Learning (DQN), Convolutional
Neural Networks (CNN), and Multi-Layer Perceptron (MLP) on
MNIST Dataset**

1. Abstract:

This project presents a comparative study of three different machine learning paradigms applied to the MNIST handwritten digit classification task: a Convolutional Neural Network (CNN), a Multi-Layer Perceptron (MLP), and a Deep Q-Network (DQN) based on Reinforcement Learning (RL). All three models were implemented from scratch using TensorFlow/Keras and evaluated on the MNIST benchmark dataset under consistent conditions. The CNN achieved the highest test accuracy of 99.26% in just 27.7 seconds of training. The MLP followed closely with 98.25% accuracy and the fastest training time of 20.4 seconds. The DQN-based RL agent achieved 93.82% accuracy but required approximately 3,580 seconds — nearly 60 times longer than the CNN. Results confirm that supervised learning methods are far more efficient for static classification tasks with labeled data, while RL provides unique advantages in sequential decision-making settings where explicit labels are unavailable.

2. Introduction

Deep learning has transformed the way computers understand visual information. Two dominant learning paradigms in modern AI are Supervised Learning and Reinforcement Learning (RL), each with distinct strengths and limitations.

In supervised learning, a model is trained on labeled examples by minimizing a loss function using gradient descent. Convolutional Neural Networks (CNNs) are the gold standard for image tasks — their convolutional filters automatically detect spatial features like edges, corners, and textures. Multi-Layer Perceptrons (MLPs) are simpler fully connected networks that treat each image as a flat vector of pixels, ignoring spatial relationships.

Reinforcement Learning, on the other hand, trains an agent to take actions in an environment by maximizing cumulative reward. The Deep Q-Network (DQN), introduced by DeepMind, combines Q-learning with deep neural networks and experience replay to achieve human-level performance in Atari games. When applied to image classification, each image is treated as a state, each predicted class as an action, and a binary reward (+1 for correct, -1 for wrong) is used instead of gradient supervision.

This project implements and evaluates all three approaches on the MNIST dataset to provide a direct, practical comparison — highlighting which paradigm is most suitable for standard image classification and why.

3. Dataset: MNIST

The MNIST (Modified National Institute of Standards and Technology) dataset is one of the most widely used benchmarks in machine learning and computer vision. It contains grayscale images of handwritten digits from 0 to 9, making it ideal for evaluating classification algorithms.

Total Images	70,000 (60,000 training + 10,000 test)
Image Dimensions	28 x 28 pixels (grayscale, single channel)
Number of Classes	10 (digits 0 through 9)
Pixel Value Range	Normalized to [0, 1] (divided by 255)
Input for CNN	Shape: (28, 28, 1) — 2D spatial format
Input for MLP & DQN	Shape: (784,) — flattened vector

The dataset is well-balanced with approximately 6,000 training images per class. Its relatively simple structure makes it an excellent testbed for comparing different learning paradigms under controlled conditions.

4. Methodology

4.1 Convolutional Neural Network (CNN)

The CNN model exploits the 2D spatial structure of images by applying learnable convolutional filters. This makes it particularly powerful for detecting local patterns such as edges, curves, and textures that are critical for digit recognition. The architecture used in this project is:

- Conv2D (32 filters, 3x3 kernel, ReLU activation) followed by MaxPooling (2x2)
- Conv2D (64 filters, 3x3 kernel, ReLU activation) followed by MaxPooling (2x2)
- Flatten layer to convert 2D feature maps into a 1D vector

- Dense layer (128 neurons, ReLU) with Dropout (0.5) for regularization
- Output layer: Dense (10 neurons, Softmax) for class probabilities

Training configuration: Adam optimizer, categorical cross-entropy loss, 10 epochs, batch size 128, 10% validation split. Total trainable parameters: 225,034.

4.2 Multi-Layer Perceptron (MLP)

The MLP is a traditional fully connected neural network. It treats each 28x28 image as a flat 784-dimensional vector, disregarding spatial relationships between pixels. While simpler than CNN, it still performs well on MNIST due to the clear pixel-level patterns in digit images.

Architecture:

- Input: Flattened image vector of size 784
- Dense (256 neurons, ReLU) with Dropout (0.3)
- Dense (128 neurons, ReLU) with Dropout (0.3)
- Output layer: Dense (10 neurons, Softmax)

Training configuration: Adam optimizer, categorical cross-entropy loss, 10 epochs, batch size 128. Total trainable parameters: 235,146. The MLP trained in just 20.4 seconds — the fastest of all three models.

4.3 Deep Q-Network (DQN) — Reinforcement Learning

The DQN agent reformulates the image classification problem as a Markov Decision Process (MDP). Instead of using labeled data and gradient-based learning, the agent learns through trial and error by receiving reward signals. The key RL concepts are mapped as follows:

State (s)	Flattened image	784-dimensional pixel vector
Action (a)	Predicted digit	One of 10 class indices (0–9)
Reward (r)	Correctness signal	+1 if correct, -1 if incorrect
Episode	Single image	Each image is one independent episode
Q-Network	Policy network	Dense(128) → Dense(64) → Dense(10, linear)

Key DQN hyperparameters: experience replay buffer of 5,000 transitions, minibatch size 64, discount factor $\gamma = 0.95$, epsilon-greedy exploration starting at 1.0 and decaying to 0.01 (decay rate 0.995 per episode), Adam optimizer with learning rate 0.001. Training was run for 10 episodes over 5,000 training samples. Replay was triggered every 50 steps within each episode.

The epsilon-greedy strategy ensures the agent starts by exploring randomly and gradually shifts to exploiting learned Q-values. By episode 10, epsilon had dropped to 0.01, meaning the agent was acting almost entirely on its learned policy.

5. Results and Comparison

5.1 Final Accuracy and Training Time

All three models were evaluated on the full MNIST test set of 10,000 images. The table below summarizes the key performance metrics:

Test Accuracy	99.26%	98.25%	93.82%
Training Time	~27.7 sec	~20.4 sec	~3580 sec
Training Samples Used	54,000	54,000	5,000
Epochs / Episodes	10 epochs	10 epochs	10 episodes
Learning Signal	Cross-entropy loss	Cross-entropy loss	Binary reward
Convergence	Fast & stable	Fast & stable	Slow, noisy initially

5.2 Per-Model Analysis

CNN: Accuracy consistently increased from roughly 90.75% (epoch 1) up to 99.16% (epoch 10). Close correlation between validation and training accuracy suggests very little overfitting, which can be attributed to dropout and pooling operations. Final test accuracy of 99.26% proves that CNN is the best model for this problem.

MLP: First epoch resulted in accuracy of 87.71%, while after epoch 10, MLP achieved 98.27% accuracy. Time required for training was also fastest, at 20.4 seconds, because of simple architecture. Although MLP doesn't take into account the spatial structure of images, MNIST digit pattern is sufficiently regular that pixels alone are enough information.

DQN (RL): Starting from 22% accuracy (first episode, random due to high epsilon), DQN managed to reach 97.68% training accuracy after 10 episodes. However, test set accuracy was just 93.82%, which indicates overfitting to the training set. Massive training time of 3,580 seconds is clearly an issue, despite using only 5,000 training samples.

5.3 Episode-wise DQN Progress

Episode	Total Reward	Train Accuracy	Epsilon	
1	-2,800	22.00%	0.6058	Random
3	+1,858	68.58%	0.2201	Learning
5	+3,682	86.82%	0.0800	Improving
7	+4,408	94.08%	0.0290	Strong
10	+4,768	97.68%	0.0100	Converged

6. Discussion

6.1 Why CNN Performs Best

CNN is best suited for image classification because it preserves and exploits the 2D spatial structure of images. Convolutional filters learn to detect local features (edges, shapes, corners) in a position-invariant manner, which is exactly what is needed to distinguish handwritten digit shapes. Pooling layers reduce spatial dimensions while retaining important features, and dropout prevents overfitting. The result is fast, stable convergence with the highest accuracy among all three models.

One important insight is that RL is not a replacement for supervised learning in tasks with available labels — it is a complementary tool for settings where explicit supervision is unavailable or the task is inherently sequential.

6.2 MLP — Simple but Competitive

Despite its lack of consideration of spatial relationships, the MLP algorithm worked very well. The reason why MLP was able to work well is that the MNIST digits were centered and standardized, with the same width for the stroke used in writing each digit. If a more complex dataset such as CIFAR-10 were used, MLP would have difficulty working effectively.

6.3 DQN — Powerful Paradigm, Wrong Task

The performance by DQN proves that RL is capable of solving classification tasks, but certainly, it cannot be considered the best algorithm in this case. The key problem here is the difference in the quality of information: RL uses binary information (-1 or +1) per each sample while supervised learning uses gradients over all the 10 classes. Thus, it takes much more time to learn. DQN had 5,000 samples and required 3,580 seconds, which is 130 times longer than CNN per sample.

RL really shines where there are several sequential decisions, delayed reward, and exploration involved – such as game playing, robotics, self-driving cars, and resource planning. In this case, there is no dataset available, and the agent learns only from environment information.

6.4 Practical Takeaways

- For image data with training labels, use CNN, as it is the most accurate and efficient method.
- MLP is suitable for simple problems involving tabular data, as well as basic image classification where speed of training is critical.
- Where the problem requires transition through states, lack of pre-existing data labels, and learning from an environment, DQN/RL becomes appropriate.
- A combined approach that involves the use of CNN to encode states within the context of a DQN algorithm is widely used today in RL (the initial DQN algorithm created by DeepMind on Atari games).

7. Conclusion

This work demonstrates the successful implementation and comparison of three different machine learning algorithms – CNN, MLP, and Reinforcement Learning based on DQN – on the MNIST handwritten digit classification problem. The main conclusions are:

- CNN model achieved the best test set accuracy rate (99.26%) and appears to be the best-performing classifier among the three because of its ability to perform spatial features extraction.
- MLP model managed to achieve 98.25% test accuracy with the fastest training time (20.4s).
- DQN achieved relatively low test accuracy (93.82%) with the training time close to 3,580 seconds.

Thus, supervised learning appears to be preferable whenever label availability is guaranteed. However, reinforcement learning is an effective method of handling sequential decision-making tasks when labels are unavailable, the problem needs to be explored, and/or the environment provides only reward feedback for each action performed.

For future research one may suggest investigating the following directions: (1) utilization of CNN as a state representation method inside the DQN agent, (2) application of RL algorithm in a more relevant environment (e.g. games or resource management), and (3) investigation of other RL algorithms, such as PPO/A3C.